Function notation

1 $f(7) = -6$ or $f : 7 \longrightarrow -6$

2

a 112 **b** 966

3

a -20 **b** 4

4

a 12 **b** -16

5

a 8 **b** -13

6

a $\frac{35}{4}$ **b** -7

7

a 68 **b** -4

8 $\frac{9}{2}$

9

a 3 **b** $\sqrt{19}$ **c** $\sqrt{4}$

10

a 0 **b** 3 **c** $\frac{91}{25}$

11

a $a^2 + 8a$ **b** $b^2 + 8b$

12

a -4 **b** $-\frac{121}{32}$

13

a 0 **b** 2.67 **c** 80.99

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14

a $\sqrt{128}$ **b** 12 **c** $\sqrt{63}$ **d** $\sqrt{142}$

15

a 12 **b** $m^2 + 8$

16

a $y = 2 - \frac{x}{3}$ **b** $f(x) = 2 - \frac{x}{3}$ **c** $f(3) = 1$

17

a $y = \frac{x}{4} - 2$ **b** $f : x \longrightarrow \frac{x}{4} - 2$ **c** $f : 12 \longrightarrow 1$

18

a $y = 3 - x - 6x^2$ **b** $f(x) = 3 - x - 6x^2$

c -54

19

a $y = 4x^2 + x + 5$ **b** $f : x \longrightarrow 4x^2 + x + 5$

c 23

20

a $y = \frac{6}{5}x + \frac{7}{5}$ **b** $f : x \longrightarrow \frac{6}{5}x + \frac{7}{5}$

c 5

21

a 2 **b** 0 **c** $x = 1$

22

a 1

b Yes, because every value of x corresponds to only one value of y.

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23

a

- Independent variable: Size of pizza
- Dependent variable: Cost of pizza

b

- Independent variable: Time spent studying
- Dependent variable: Mark achieved on a test

c

- Independent variable: Amount borrowed
- Dependent variable: Duration of a loan

d

- Independent variable: Time spent exercising
- Dependent variable: Physical fitness level

24

a **i** -48 **ii** 15 **iii** -49 **b** $x = 7.81, x = -7.81$

25

a $ak^3 - 3k + 5$ **b** $-ak^3 + 3k + 5$ **c** No **d** No

Applications

26 $y = -7, -5$

27

a 320 **b** $x = -4, x = 2, x = -2$

28

a $c = -7$ **b** $a = b - 1$ **c** $21 = 4a + b$ **d** $a = 4, b = 5$

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29

a $P(x) = -\frac{x^2}{4} + 35x - 410$

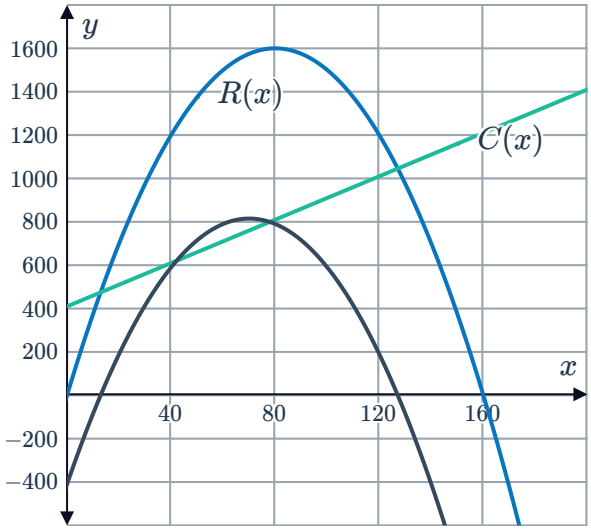
b

i $R(70) = 1575,$

ii $C(70) = 760$

iii $P(70) = 815$

c



30

a $T(x) = x^2 - 182x + 25650$

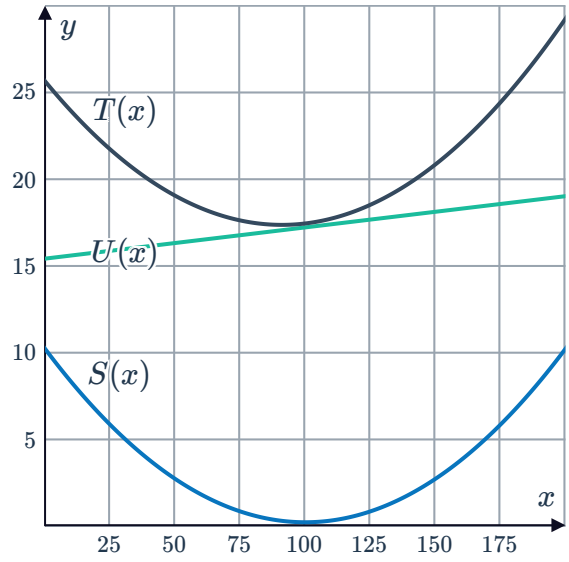
b

i $S(91) = 308$

ii $U(91) = 17061$

iii $T(91) = 17369$

c



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